Talank Baral

Ph.D. Student in Computer Science
Department of Computer Science, George Mason University • Falls Church, VA 22046, USA
tbaral@gmu.edu talank.com.np github.com/talank in/tlnkb

Summary

Software engineer and Ph.D. student in Computer Science, available for full-time work now, focused on testing, continuous integration, and build reliability. Over two years in industry building and maintaining automated end-to-end, API, and GUI test suites and Docker-based CI pipelines for ownCloud, an open-source file-sync and collaboration platform, followed by four years of research building tools that make CI builds faster and tests more reliable.

Education

2022–Present **Ph.D. in Computer Science** — Fairfax, VA, USA
George Mason University
GPA: 3.98/4.00 • Advisor: Wing Lam

2015–2019 **B.E. in Software Engineering** — Pokhara, Nepal
Gandaki College of Engineering and Science, Pokhara University
GPA: 3.67/4.00

Experience

Aug 2022–Present **Graduate Research/Teaching Assistant** — Fairfax, VA, USA
Department of Computer Science, George Mason University

- Reproduced timing-dependent flaky test failures with a neurosymbolic LLM pipeline (TDRepro; ASE ’26 Distinguished Paper)
- Detected and prevented unnecessary content generation in CI builds (OptCD; ASE ’23 and ICSE ’25 Demo)
- Studied the test coverage of pull requests authored by AI coding agents (ICSME ’26)
- Developed techniques for efficient test orders in regression testing

Jan 2020–Aug 2022 **Software Developer** — Pokhara, Nepal
JankariTech

- Built and maintained BDD (Gherkin) acceptance test suites for ownCloud, an open-source file-sync and collaboration platform: web UI (Nightwatch.js) and the desktop sync client (Squish GUI tests)
- Made CI failures diagnosable: HTTP and server logs, desktop screenshots on test failure, stack traces for crashed applications under test, and a parser for GUI test logs
- Maintained Drone CI pipelines, including nightly cron runs with chat notifications and Dockerized test middleware, and kept integration test suites in sync across ownCloud server, oCIS, and Reva
- Refactored tests to the Page Object Model and fixed intermittent failures and tests that passed when they should have failed
- Set up test automation, CI pipelines, and acceptance tests for client projects

Selected Projects

2023–2025 **OptCD** [Code] [Demo]
Python, Bash, GitHub Actions, Maven, Gemini API

- Instruments a GitHub Actions workflow to find directories generated but never accessed, traces

each to the responsible Maven plugin and step, and uses an LLM (Gemini) to propose a fixed workflow that it pushes for validation in CI
- Built the evaluation pipeline: mined Java repositories through the GitHub API, filtered for Maven and GitHub Actions projects, and aggregated file-access logs from 600+ CI job runs into per-job and per-project results
- Findings led to 25 pull requests to open-source projects that disable unnecessary generation

2016–2021 **Chamena Griha** [Code]

PHP, MySQL, JavaScript, Nightwatch.js, Cucumber, Selenium, Docker, Travis CI

- Food-ordering web app for college canteens with separate student, canteen, and college-admin roles
- Added Gherkin acceptance tests (Nightwatch.js + Cucumber) and a Travis CI pipeline that provisions MySQL, seeds the schema, and runs Selenium in Docker

2018–2019 **Gas Station Management System** [Code] [Demo]

Java, JavaFX, Apache Derby, Maven

- Desktop point-of-sale and accounting app for a fuel station built with JavaFX and an embedded Apache Derby database
- Covers instant and credit billing, credit-company accounts, fuel stock and tank refills, daily closing reports, and receipt printing

Technical Skills

Languages	Python, Java, JavaScript, Bash, SQL, PHP, C, C++
Testing	JUnit, Selenium, Cypress, Nightwatch.js, Jest, Puppeteer, Behat, Squish, Cucumber (BDD/Gherkin)
CI/CD	GitHub Actions, GitLab CI, Travis CI, Drone, Docker, Maven, Git
Data & APIs	MySQL, Apache Derby, GitHub REST API, Gemini API

Open-Source Contributions

- **International Dataset of Flaky Tests (IDoFT)**: Detected and fixed flaky tests in open-source Java projects
- **CI build optimization**: Submitted 25 pull requests to open-source projects that disable generation of unnecessary content in CI builds
- **ownCloud**: Automated tests for the ownCloud web and desktop clients
- **pact-cplusplus**: Fixed compilation issues in version 0.1.0

Publications

Conference Papers

[C3] ASE '26

Shanto Rahman, **Talank Baral**, August Shi, and Wing Lam. **TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures**. *41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*, pp. 675–686, Munich, Germany, 2026.

ACM SIGSOFT Distinguished Paper Award

Acceptance rate: 8.82% direct acceptance [DOI] [PDF]

[C2] ICSME '26

Atish Kumar Dipongkor, **Talank Baral**, Wing Lam, and Kevin Moran. **Test Coverage Analysis of Agentic Pull Requests**. *42nd International Conference on Software Maintenance and Evolution (ICSME 2026)*, Vision and Emerging Results Track, Benevento, Italy, 2026.

Acceptance rate: 39% [arXiv] [Code] [Artifact]

[C1] ASE '23 — **Talank Baral**, Shanto Rahman, Bala Naren Chanumolu, Basak Balci, Tuna Tuncer, August Shi, and Wing Lam. **Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation**. *38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*, pp. 901–913, Kirchberg, Luxembourg, 2023.
Acceptance rate: 21% (134/630) [PDF] [Slides]

Tool Demonstrations

[D1] ICSE '25 Demo — **Talank Baral**, Emirhan Oğul, Shanto Rahman, August Shi, and Wing Lam. **OptCD: Optimizing Continuous Development**. *47th IEEE/ACM International Conference on Software Engineering (ICSE 2025)*, Demonstrations Track, pp. 45–48, Ottawa, ON, Canada, 2025.
[PDF] [Slides]

Under Review

[U1] **Talank Baral**, Abdelrahman Baz, Atish Kumar Dipongkor, August Shi, Kevin Moran, and Wing Lam. **Using Efficient Test Orders for Regression Testing**. *Under review*.

Honors & Awards

Oct 2026 — **ACM SIGSOFT Distinguished Paper Award**, ASE '26
TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures

Aug 2023 — **SIGSOFT CAPS Travel Grant**, ACM SIGSOFT
Awarded $1,200 to attend ASE 2023 (declined due to visa constraints)

May 2023 — **Summer Research Award**, Department of Computer Science, George Mason University

2015–2019 — **Undergraduate Academic Excellence Award**, Gandaki College of Engineering and Science, Pokhara University
Merit-based award

Feb 2018 — **First Place, Project Expo**, 7th GCES IT Expo, Pokhara University

Jun 2012 — **National Talent Identification Award (First Place Nationwide)**, Government of Nepal
For designing a seed plantation machine

Teaching

Fall 2026	**SWE 637 / SWE 437**: Software Testing	*Graduate Teaching Assistant*
Spring 2026	**CS 691**: Synergies of LLMs and Software Engineering	*Graduate Teaching Assistant*
Fall 2022	**CS 262**: Introduction to Low-Level Programming	*Graduate Teaching Assistant*

Invited & Conference Talks

May 2025 — *OptCD: Optimizing Continuous Development*
ICSE 2025 (Demonstrations Track) • Conference Talk • Ottawa, ON, Canada

Jan 2024 — *Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation*
ESSEC 2024 • Colloquium Talk • William & Mary, Williamsburg, VA, USA

Jul 2023 — *Optimizing Continuous Development*
Microsoft, Redmond Campus • Industry Talk • Redmond, WA, USA

Jul 2023 — *Optimizing Continuous Development and Predicting Test Information in Software Development*
ISSTA 2023 • Doctoral Symposium • Seattle, WA, USA